🧪 Step-by-Step Lab: Redshift + S3 + Analytics

🧰 Prerequisites

- AWS account with admin permissions
- Sample CSV dataset (e.g., `users.csv`)
- IAM Role with S3 read access for Redshift

🔧 Step 1: Create an S3 Bucket and Upload Dataset

1. Go to **S3 > Create bucket** (e.g., `redshift-lab-bucket`)
2. Upload `users.csv` to the bucket

🏗️ Step 2: Create an IAM Role

1. Go to **IAM > Roles > Create Role**
2. Choose **Redshift** as the use case
3. Attach the **AmazonS3ReadOnlyAccess** policy
4. Copy the **Role ARN**

🏗️ Step 3: Launch a Redshift Cluster

1. Go to **Amazon Redshift > Create Cluster**
2. Choose:
    - Node type: `dc2.large` (Free tier eligible)
    - Database name: `dev`

    - Master user: `admin`, Password: `Password123!`

3. Associate the IAM Role created earlier

4. Enable **Publicly Accessible**

🔌 Step 4: Connect to Redshift

- Use **Query Editor V2** in AWS Console or connect via **DBeaver/SQL Workbench/J**

🗃️ Step 5: Create Table

```
CREATE TABLE users (
    userid INT,
    username VARCHAR(50),
    age INT,
    gender CHAR(1),
    city VARCHAR(50)
);
```

📥 Step 6: Load Data from S3

```
COPY users
FROM 's3://redshift-lab-bucket/users.csv'
IAM_ROLE 'arn:aws:iam::123456789012:role/MyRedshiftRole'
CSV IGNOREHEADER 1;
```

📊 Step 7: Run Analytics Queries

```
SELECT gender, COUNT(*) FROM users GROUP BY gender;
SELECT city, AVG(age) FROM users GROUP BY city ORDER BY AVG(age) DESC;
```
